

HWA CHONG INSTITUTION

C2 UWA 2025

COMPUTING

MAR 2025

Paper 2 (9569 / 02)

1 hour

Additional Materials:

Electronic version of STAFF . j s o n data file

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

All tasks must be done in the computer laboratory. You are not allowed to bring in or take out any pieces of work or materials on paper or electronic media or in any other form.

Approved calculators are allowed.

Save each task as it is completed.

The use of built-in functions, where appropriate, is allowed for this paper unless stated otherwise.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is **20**.

1. Design the server program and client program for a nugget game.

Sarah and Chloe bought a large number of chicken nuggets. They took turns to take 1 to 3 nuggets, and agreed that the person who took the last piece would have to pay for all the nuggets.

The game can be simulated with two programs, one for Sarah and one for Chloe such that:

- Sarah starts by inputting the number of nuggets available and sends this number to Chloe.
- Chloe starts taking the nuggets by inputting a number.
- Sarah and Chloe then alternate in taking nuggets each:
 - Each girl can take exactly 1, 2 or 3 nuggets each turn.
 - Each girl cannot take more than the number of nuggets left.
 - Prompt to ask again if an invalid number is provided.
- After nuggets are taken by either girl:
 - If there are nuggets left, this number is sent over.
 - If no nuggets are left after taking:
 - 'end' is sent
 - 'You lose.' is displayed.
- 'You win.' is displayed if the opponent takes the last nugget.

The following is an example of what each girl would see.

<u>Sarah</u>	<u>Chloe</u>
How many nuggets? 11	There are 11 nuggets left. How many to take? 3 3 nuggets taken, 8 left.
There are 8 nuggets left. How many to take? 4 There are 8 nuggets left. How many to take? 2 2 nuggets taken, 6 left.	There are 6 nuggets left. How many to take? 0 There are 6 nuggets left. How many to take? 2 2 nuggets taken, 4 left.
There are 4 nuggets left. How many to take? 2 2 nuggets taken, 2 left.	There are 2 nuggets left. How many to take? 3 There are 2 nuggets left. How many to take? 1 1 nuggets taken, 1 left.
There are 1 nuggets left. How many to take? 1 You lose.	You win.

The socket protocol uses `\n` to detect the end of a message.

Save your program codes as `TASK1_SARAH_<your name>_<ct>.ipynb` and `TASK1_CHLOE_<your name>_<ct>.ipynb` [10]

2. Name your Jupyter Notebook as `TASK2_<your name>_<ct>.ipynb`

The task is to create a NoSQL database for a company to manage its staff records on the recent vaccine. The vaccine requires two doses to be completed.

Task 2.1

Write a python program to:

- Create a MongoDB database named `Vaccine` and a new collection named `Staff`.
- Insert the documents into the `Staff` collection using the data file `STAFF.json`. [2]

Task 2.2

Write program code which make use of `Staff` collection in `Vaccine` database to:

- display the `Name`, `Age` and `Reason` for staff who do not accept to take the vaccine;
- display the number of staff with at least one dose of vaccine taken;
- add Charlie Lee's second dose on "5 Apr 2021" and display his updated record;
- display the name(s) of staff with at least 50 years old and completes both doses

All outputs should have appropriate messages to indicate what you are showing.

Save your Jupyter Notebook for Task 2. [8]

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